MOHAMMED SIDDIQUE

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Portfolio: https://mohammed-siddique-data-ai-portfolio.vercel.app
GitHub: https://github.com/MohammedSiddique8998
Portfolio Repository: https://github.com/MohammedSiddique8998/mohammed-siddique-data-ai-portfolio

PROFESSIONAL PROFILE

MSc Data Science and Artificial Intelligence student at the University of Liverpool with over five years of professional experience across data validation, catalogue management, automation workflows, systems configuration, CAD/design-to-production processes, client support and process improvement. Strong foundation in Python, SQL, Power BI, Excel, machine learning, data cleaning, exploratory analysis, statistical thinking and applied AI. Experienced in working with structured datasets, identifying data quality issues, performing root cause analysis, documenting workflows, and translating business requirements into reliable technical outputs. Able to communicate with technical and non-technical stakeholders, support cross-functional teams, and connect analytical problem-solving with practical business impact. Seeking data, analytics, data science, AI engineering or AI consulting roles, including internships and entry-level industry placements.

CORE SKILLS

- Data analysis, data cleaning, exploratory data analysis and feature engineering.
- SQL querying, data validation, transformations and data quality checks.
- Python-based analysis using Pandas, NumPy, Scikit-learn, Matplotlib and Seaborn.
- Machine learning model development, classification, regression, model evaluation and experiment comparison.
- Data visualisation, dashboards, KPI reporting and stakeholder-ready insight generation.
- Process automation, rule-based logic, workflow optimisation and operational troubleshooting.
- Requirements gathering, stakeholder communication, documentation and user support.
- Root cause analysis, 5 Whys, SWOT analysis, issue triage and continuous improvement.
- Business analysis, problem definition, solution design and practical AI opportunity assessment.
- Cross-functional collaboration with engineering, product, operations and client-facing teams.

TECHNICAL SKILLS

Programming and Querying: Python, SQL, basic C

Data Analysis and Statistics: Pandas, NumPy, data cleaning, data preprocessing, EDA, feature engineering, descriptive analytics, pattern detection, applied statistics, hypothesis-testing exposure
Machine Learning and AI: Scikit-learn, supervised and unsupervised learning, logistic regression, random forests, support vector machines, neural networks, classification models, regression models, model evaluation, confusion matrices, reinforcement learning concepts

Deep Learning: TensorFlow and Keras project exposure; PyTorch learning exposure; CNNs, transfer learning, VGG16 and ResNet50 exposure through current project work

Visualisation and BI: Power BI, Matplotlib, Seaborn, Excel dashboards and reporting

Forecasting and Time Series: time-based feature engineering, trend and seasonality analysis, moving averages, regression-based forecasting, MAE/RMSE model evaluation exposure

Data Collection and Automation: API integration learning, REST APIs learning, JSON handling, web data extraction with Requests and BeautifulSoup learning, data pipelines and rule-based automation

Tools and Platforms: Jupyter Notebook, Google Colab, VS Code, Anaconda, Git basic, Microsoft Excel, Word, PowerPoint, AutoCAD, Revit, MS Office

EDUCATION

MSc Data Science and Artificial Intelligence with Year in Industry
University of Liverpool, United Kingdom | September 2025 - Present

- Relevant modules include Programming Fundamentals for Data Science, Databases and SQL, Data Mining and Visualisation, Machine Learning and Bio-Inspired Learning, Computational Intelligence, Applied AI and Machine Learning, Applied Statistics, Research Methods and Employability Skills.
- Developed practical experience in Python programming, NumPy, Pandas, Scikit-learn, SQL, data visualisation, model experimentation, analytical problem-solving and applied AI research.

Bachelor of Engineering, Civil Engineering
Visvesvaraya Technological University / KNS Institute of Technology, Bengaluru, India | 2017 - 2020

- CGPA: 8.19

Diploma in Civil Environmental Engineering
Sri Jayachamarajendra Government Polytechnic, Karnataka, India | 2014 - 2017

- Distinction

PROFESSIONAL EXPERIENCE

AI Consulting Micro-Internship
Accenture, UK | November 2025 - December 2025

- Analysed business problems and identified opportunities for AI-driven solutions.

- Applied consulting frameworks to structure problems, assess process inefficiencies and propose practical automation opportunities.
- Designed an AI solution concept for proposal-generation workflows, combining NLP-based information extraction with rule-based structuring and prioritisation.
- Communicated technical ideas in business-focused language for non-technical stakeholders.
- Strengthened client-focused thinking, problem definition, solution evaluation and professional reporting skills.

Catalogue Executive

Infurnia Pvt Ltd, India | December 2023 - date to confirm

- Managed and optimised structured catalogue/configuration datasets across 150+ completed projects, improving data accuracy and reducing inconsistencies in automated outputs.
- Performed data validation, SKU mapping, product-parameter checks and rule-based logic checks to ensure reliable system outputs.
- Analysed configuration and workflow datasets to identify recurring inconsistencies, incorrect mappings and output-quality issues.
- Collaborated with engineering, product and operations teams to resolve automation issues, refine validation logic and improve system reliability.
- Investigated configuration problems using structured analysis and root cause thinking, contributing to smoother workflows and reduced manual correction.
- Supported reporting, documentation and knowledge-sharing around data processes, validation rules and configuration logic.
- Worked with data-driven systems requiring accuracy, testing, quality assurance and production-ready outputs.

Pre-Production Engineer

Finemake Pvt Ltd, India | 2023

- Cleaned, transformed and analysed production-related datasets to support material usage, reporting accuracy and workflow efficiency.
- Translated design specifications into structured datasets for accurate and automated pre-production workflows.
- Generated and validated production outputs, helping reduce errors and improve operational reliability.
- Partnered with technical and production teams to improve data models, reporting accuracy and process performance.

Nucleus Configuration Specialist / IMOS Designer

Homelane Pvt Ltd, India | December 2021 - June 2023

- Configured backend automation logic for 200+ workflow processes, reducing manual intervention and improving operational efficiency.
- Worked with configuration and pricing data pipelines supporting design-to-production systems.
- Conducted root cause analysis to diagnose data inconsistencies, workflow bottlenecks and recurring output errors.
- Validated large datasets and rule-based configurations to maintain high standards of data integrity and system reliability.
- Analysed operational datasets to identify inefficiencies and support workflow optimisation.
- Collaborated with design, engineering, operations and product teams to improve data-driven workflows and troubleshoot system issues.
- Supported testing of new features, data models and system changes prior to release.

AutoCAD Draftsman / Project Coordinator / Trainee

Kooheji Contractors W.L.L, Bahrain; H2S Constructions, India | dates to confirm

- Worked across civil, joinery and design-related projects, using structured technical data, drawings and documentation to support delivery.
- Applied analytical and problem-solving skills to coordinate project information from design through execution.
- Used CAD tools and structured documentation to maintain accurate technical records and support compliance with project requirements.
- Coordinated with engineers, clients and stakeholders to clarify requirements, track progress and support timely delivery.
- Used data-driven tracking and documentation to monitor progress, organise resources and improve workflow efficiency.

DATA SCIENCE / AI / ANALYTICS PROJECTS

Mental Health Risk Prediction using Machine Learning

Academic Project, University of Liverpool

- Developed classification models using Logistic Regression, Random Forest and Support Vector Machines to predict mental health risk from survey datasets.
- Performed data preprocessing, missing-value handling, outlier detection, normalisation, feature scaling and data preparation using Pandas and NumPy.
- Conducted exploratory data analysis to identify patterns, relationships and drivers associated with mental health outcomes.
- Evaluated model performance using accuracy metrics and confusion matrix visualisation, comparing trade-offs between performance and interpretability.

Applied AI Chest X-ray Image Classification

Portfolio Project

- Built a cleaned computer vision portfolio project for three-class chest X-ray classification: Normal, Opacity and Pneumonia.
- Developed a reproducible PyTorch training and evaluation pipeline with a custom CNN and DenseNet121 feature extraction/fine-tuning options.
- Documented seed, batch size, image size, split assumptions, preprocessing, augmentation, class imbalance handling and model selection rules.

- Added imbalance-aware evaluation outputs including balanced accuracy, macro precision/recall/F1, weighted F1, class-wise reporting and normalised confusion matrix generation.
- Corrected the public repository to remove unsupported historical result claims; metrics are intended to be regenerated locally using a permitted dataset and current code.
- GitHub repository: https://github.com/MohammedSiddique8998/applied-ai-image-classification

AI-Driven Proposal Automation System
AI Consulting Project, Accenture Micro-Internship / University of Liverpool, 2025

- Designed an NLP-based automation solution to extract project scope details and support automated proposal generation.
- Analysed workflow inefficiencies using 5 Whys and SWOT analysis to identify opportunities for AI-enabled automation.
- Proposed a scalable AI system with rule-based scoring and structured outputs to reduce ambiguity and improve proposal consistency.
- Estimated potential to reduce proposal preparation turnaround time by 40-60%, based on the proposed automation design.

AI and Machine Learning Techniques for Legal Judgement Prediction in Criminal Law
University of Liverpool Academic Project, 2025

- Reviewed machine learning algorithms including SVMs, Random Forests and Neural Networks for legal judgement prediction tasks.
- Evaluated feature engineering approaches, model-performance considerations, interpretability constraints and dataset limitations in a high-stakes legal context.
- Identified future development areas including explainable AI, fairness-aware modelling and data augmentation.

Bio-Inspired Multi-Armed Bandit Reinforcement Learning

Portfolio Project

- Built a clean NumPy-based reinforcement learning simulation for stationary Gaussian n-armed bandits across 5-arm, 10-arm and 20-arm settings.
- Compared random, greedy, epsilon-greedy and Upper Confidence Bound policies to analyse exploration-exploitation trade-offs.
- Generated reproducible average-reward and optimal-action-rate plots from current code using 500 runs and 1,000 steps per experiment setting.
- Summarised final-window policy performance, with UCB c=2 producing the strongest average reward in the generated experiments.
- GitHub repository: https://github.com/MohammedSiddique8998/bio-inspired-multi-armed-bandit-rl

Data Mining Text Clustering and Visualisation

Portfolio Project

- Built a public-safe unsupervised text clustering project using sentence preprocessing, TF-IDF vectorisation, K-Means clustering and PCA visualisation.
- Created a reproducible demo corpus and generated current results from code, including selected K=4, silhouette score of 0.2111 and balanced 10-document clusters.
- Interpreted clusters using top TF-IDF terms across data analytics, healthcare, energy/sustainability and career/portfolio themes.
- Documented dataset limitations and provided instructions for replacing the demo corpus with a permitted local TSV dataset.
- GitHub repository: https://github.com/MohammedSiddique8998/data-mining-text-clustering

Energy Demand Forecasting using Time-Series Analysis
Self Project, Ongoing

- Developed an initial time-series forecasting workflow using publicly available electricity demand and/or weather datasets.
- Performed preprocessing including missing-value handling, feature scaling and time-based feature extraction such as hour, day and seasonality.
- Applied regression-based forecasting and moving-average techniques to model trends and short-term demand patterns.
- Visualised actual versus predicted demand using Matplotlib and Seaborn, and evaluated model performance using MAE/RMSE.
- Currently exploring ARIMA and machine learning-based forecasting methods to improve predictive performance.

Applied Data Analytics Workflows
Academic and Personal Project Work

- Built end-to-end analysis workflows involving SQL-style extraction, Python transformation, dashboard-oriented visualisation and reporting.
- Practised preparing raw data for analysis, validating assumptions and presenting findings through clear reports and visuals.

Interactive Data Science and AI Portfolio Website

Personal Portfolio Project

- Built and deployed a responsive portfolio website for data science, AI engineering, analytics and AI consulting roles.
- Implemented filterable project cards, case-study modals, CV downloads, contact workflows, animated visuals and a guided enquiry form using static HTML, CSS, JavaScript, Canvas and WebGL.
- Published the source code on GitHub and deployed the live site on Vercel.
- Live portfolio: https://mohammed-siddique-data-ai-portfolio.vercel.app
- GitHub repository: https://github.com/MohammedSiddique8998/mohammed-siddique-data-ai-portfolio

CERTIFICATIONS AND TRAINING

- MITx: 6.86x Machine Learning with Python - From Linear Models to Deep Learning
- PGP in Data Science and Machine Learning

- MSSQL Developer Course
- Power BI Training
- Python for Data Science Training
- Duolingo English Test
- AI Consulting Micro-Internship, Accenture UK

ADDITIONAL SKILLS AND ACHIEVEMENTS

- International work exposure across India, Bahrain and the UK.
- Experience supporting clients, gathering feedback, documenting processes and acting as a bridge between users and technical teams.
- Strong communication, active listening and peer-support skills, including informal CV, application and interview guidance for peers and junior students.
- Languages: English, Hindi, Kannada and Urdu.
- Interests include applied AI, data-driven problem solving, emerging technologies, sustainability/energy analytics and community mentoring.

CONTENT REVIEW NOTE

This master CV combines the strongest supported content from the previous CVs, tailored data/AI CVs, application forms, cover letters and project descriptions. It consolidates the MSc Data Science and Artificial Intelligence background, technical skills, professional experience in catalogue management, configuration, automation, data validation, CAD/project coordination, customer support, and the main data science/AI projects into one ATS-friendly UK CV structure.

Areas to manually confirm before using:

- Infurnia date range appears as December 2023 - November 2024 in some files and December 2023 - August 2025 in others.
- Kooheji Contractors / Bahrain role dates should be confirmed, especially because one CV indicates November 2024 - August 2025.
- Confirm whether "Customer Success Manager" or "Catalogue Executive" is the preferred official Infurnia title, as both appear in the documents.
- Confirm whether TensorFlow, Keras, PyTorch, APIs and web scraping should remain labelled as "learning/project exposure" or upgraded after further completed work.